A Supplementary Results

This appendix provides the experimental results omitted from Section 5 because of space limitations.

Figure 7 reports the search performance on SIFT1M, Deep1M, and SPACEV1M on static data. The results are consistent with those on the corresponding 10M datasets in Section 5.1, indicating scalable search performance to dataset sizes.

Table 7 reports the construction time and memory consumption on the 10M datasets. Compared with the 1M results in Section 5.1, both construction time and memory consumption of Ours scale linearly to dataset size.

Figures 8 and 9 report the results for $k = 20$ and 100, respectively. Although the search performance degrades only slightly for larger k, the overall trend does not change, and Ours consistently achieves high search performance.

Figures 10 and 11 report the search performance results on Deep1M and SPACEV1M after insertions and deletions, respectively. These results are consistent with those on SIFT1M and YouTube1M reported in Section 5.2.

Table 7: Construction time and memory consumption on 10M datasets

Dataset	Method	Const. time [sec]	Mem. consumption [GB]
Deep10M	DIGRA	62,548	17.52
	DSG	5,615	59.87
	WoW	2,814	9.42
	Ours	**1,667**	**26.82**
SIFT10M	DIGRA	69,803	17.52
	DSG	5,434	61.83
	WoW	3,113	9.42
	Ours	**1,904**	**28.01**
SPACEV10M	DIGRA	71,444	17.52
	DSG	5,103	60.02
	WoW	2,857	9.42
	Ours	**2,056**	**27.11**

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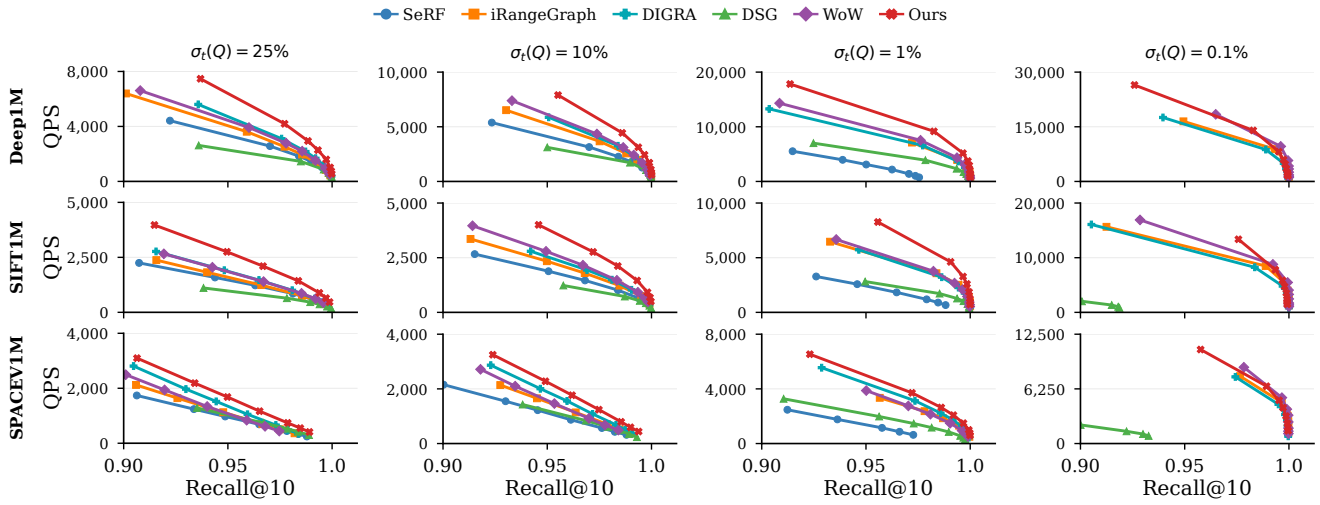


Figure 7: QPS-recall trade-off on static setting

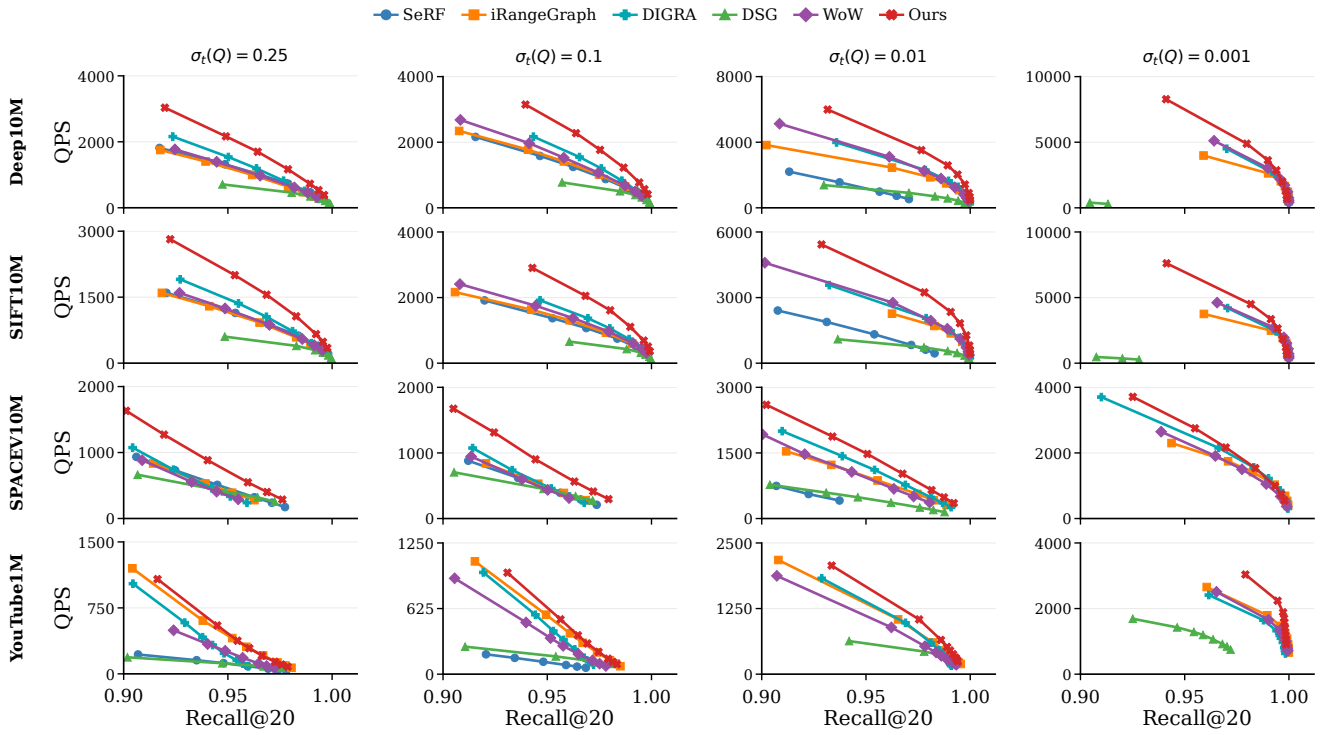


Figure 8: QPS-recall trade-off on static setting ($k = 20$)

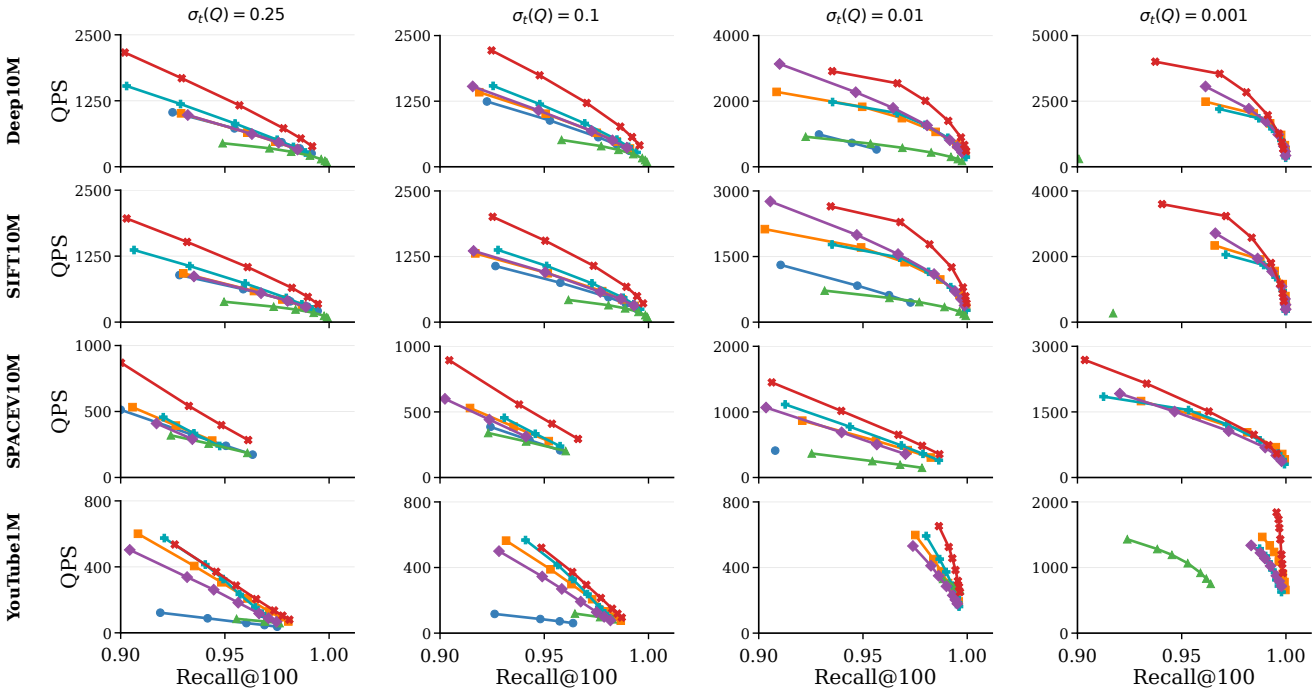


Figure 9: QPS-recall trade-off on static setting ($k = 100$)

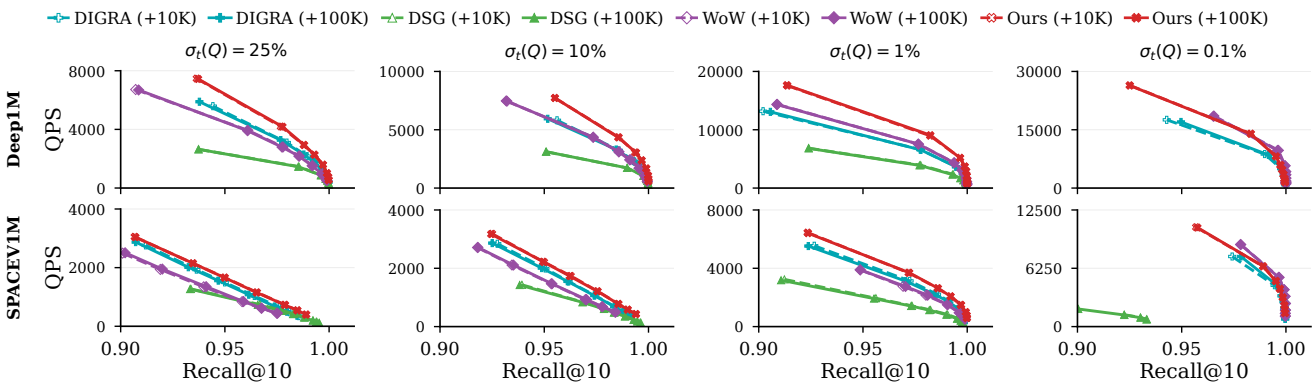


Figure 10: QPS-Recall trade-off after 10K and 100K insertions on Deep1M and SPACEV1M

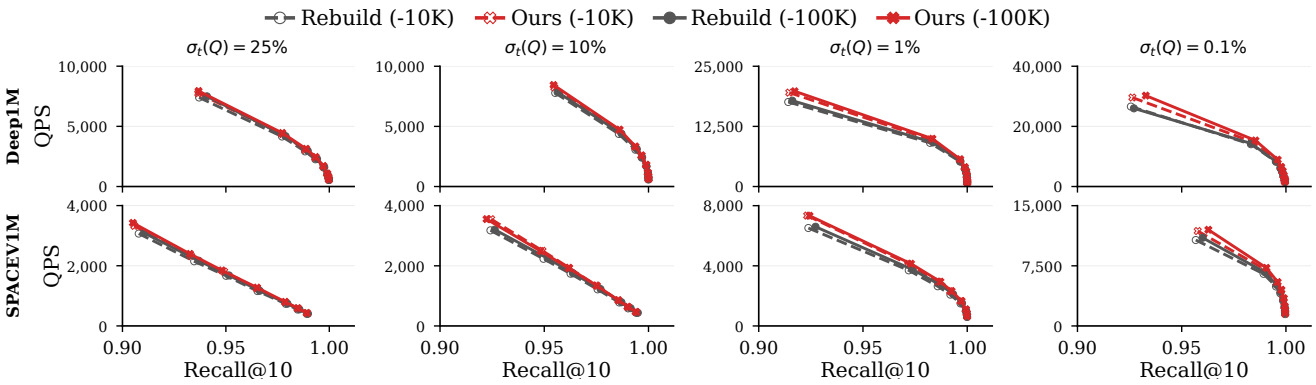


Figure 11: QPS–Recall trade-offs after 10K and 100K deletions on Deep1M and SPACEV1M